Anna's Example Thesis

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The Head
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Dear Sir,

I submit this thesis entitled "*Anna's Example Thesis*", based on MXEN4005 Mechatronics Engineering Research Project 1 and MXEN4006 Mechatronics Engineering Research Project 2, undertaken by me as part-requirement for the degree of B. Eng. Mechatronic Engineering.

Yours faithfully,

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1 Acknowledgements

I would like to acknowledge and thank the contribution and support of , my Thesis Supervisor and .

2 Abstract

Mechanical wear and fatigue failure is one of the most significant costs in industrial nations.

3 Nomenclature

Symbol	Title	Unit(s)
a	Half Contact Width	mm
b	Effective Face Width	mm

Contents

List of Figures

4 Introduction

4.1 intro stuff

The fatigue of spur gears describes the longevity of the gears in terms of the cyclical use of these components. Understanding the fatigue of these mechanical components allows for a better understanding of the expected life span of these components and advises on service intervals and replacement periods. These gears can be used in large industrial components, and if failure occurs it could not only result in the system no longer functioning but also cease production, which on large scale projects such as the mines of Australia, can result in millions of dollars in lost revenues due to an unplanned shut down. Hence, an improved understanding of the likelihood of the failure of spur gears will allow for more useful inspections of these components.

5 Background

5.1 Cost of Wear and Fatigue

As stated in Fortescue's Annual 2024 Report, they had an annual revenue of

6 Experimental Procedure

6.1 Construction of temperature distribution model

When constructing the thermal model, the body temperature will be assumed to be between

7 Results and Discussion

7.1 Construction of Thermal distribution

Figures 13 and 14 show the thermal distribution over the FGZ-A Gear and Pinion. Where the areas highlighted in red show 200°C and the dark blue area over the rings shows 120°C, as depicted in Figure 15.

8 Conclusion

These simulations and models were able to validate an appropriate method to incorporate the thermal effects expected in a gearbox

9 Future Work

9.1 Analytical Validation of Control Simulation

This Analysis shows how to validate a control model with no thermal effects or pitting; this method can be used

10 Fun Stuff

this is loads of fun

References

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11 Appendix A

STUFF IN APPENDIX A

12 Appendix B

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